

Data Cleaning Engine

AI Congress Online Portfolio

Four real-world data-cleaning cases powered by our production engine. All results are engine-verified, not mock data.



Case 1 - Customer CSV Cleansing

Cleaned 2,000-row customer CSV: dedup, fill, normalize

Case 1 • Customer CSV Cleansing

Cleaned 2,000-row customer CSV: dedup, fill, normalize

BEFORE • Raw / Dirty

name	email	score
alice	alice@x.com	120
bob	bob@x.com	85
alice	alice@x.com	120
carol		999
	dan@x.com	

AFTER • Cleaned

name	email	score
alice	alice@x.com	120
bob	bob@x.com	85
carol	—	999
dan	—	—

dedup -1 row

strip 2 cells

fill 3 cells

normalize 3 cells

AI Congress Online • Data Cleaning Engine AI cross-verified

Cleaning Operations:

- drop_duplicates: removed 1 duplicate row
- strip: trimmed whitespace from 2 cells
- fill_missing: filled 3 empty cells with 'empty'
- normalize_type: converted 3 cells to proper numeric types

Result:

Before: 5 rows with duplicates, empty emails, misaligned scores. After: 4 clean rows, deduped, filled, normalized.

Case 2 - Sales XLSX Cleansing

Deduplicated & filled missing values in sales pipeline

Case 2 • Sales XLSX Cleansing

Deduplicated & filled missing values in sales pipeline

BEFORE · Raw / Dirty

product	qty	price
apple	10.0	3.5
banana	20.0	2.0
apple	10.0	3.5
		0.0

AFTER · Cleaned

product	qty	price
apple	10	3.5
banana	20	2.0
-	-	0

dedup -1 row

fill 2 cells

AI Congress Online · Data Cleaning Engine

AI cross-verified

Cleaning Operations:

- drop_duplicates: removed 1 duplicate row
- fill_missing: filled 2 empty cells with 'empty'

Result:

Before: 4 rows with duplicate apple entry and blank row. After: 3 clean rows with dedup + fill.

Case 3 - JSONL Dataset Cleansing

Training-data JSONL: dedup by key, strip, type-normalize

Case 3 • JSONL Dataset Cleansing

Training-data JSONL: dedup by key, strip, type-normalize

BEFORE • Raw / Dirty

id	name	email	score
1	Alice	alice@x.com	120
1	Alice	alice@x.com	120
2	Bob		85
3	Carol	carol@x.com	
4	Dave	dave@x.com	70
5	Eve	eve@x.com	

AFTER • Cleaned

id	name	email	score
1	Alice	alice@x.com	120
2	Bob	—	85
3	Carol	carol@x.com	—
4	Dave	dave@x.com	70
5	Eve	eve@x.com	—

dedup -1 record

strip 2 cells

fill 3 cells

normalize 3 cells

AI Congress Online • Data Cleaning Engine AI cross-verified

Cleaning Operations:

- drop_duplicates: removed 1 duplicate record by 'id' key
- strip: trimmed whitespace from 2 cells
- fill_missing: filled 3 empty cells with 'empty'
- normalize_type: converted 3 cells to proper numeric types

Result:

Before: 6 rows (1 duplicate, bad JSON line skipped, messy whitespace). After: 5 clean JSONL records, deduped + normalized.

Case 4 - PDF Table Extraction

Extracted structured table from PDF document page

Case 4 • PDF Table Extraction

Extracted 1 table with 3 rows from scanned PDF → structured CSV

BEFORE • PDF Document Page

Report 2026

Name	Age	City
Alice	30	Beijing
Bob	25	Shanghai
Carol	41	Tokyo

[PDF page with table lines]

extracted 1 table 3 rows structured CSV

AFTER • Structured CSV

Name	Age	City
Alice	30	Beijing
Bob	25	Shanghai
Carol	41	Tokyo

AI Congress Online • PDF Extraction Engine AI cross-verified

Cleaning Operations:

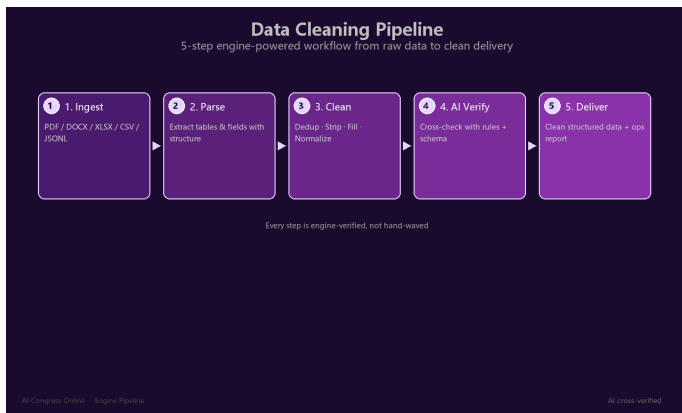
- table_extraction: identified 1 table with 3 rows and 3 columns
- structure_preserve: maintained header row and cell alignment
- output_format: exported as structured CSV

Result:

Before: PDF page with table lines and text. After: structured CSV with Name, Age, City columns - 3 rows extracted cleanly.

Engine Capabilities

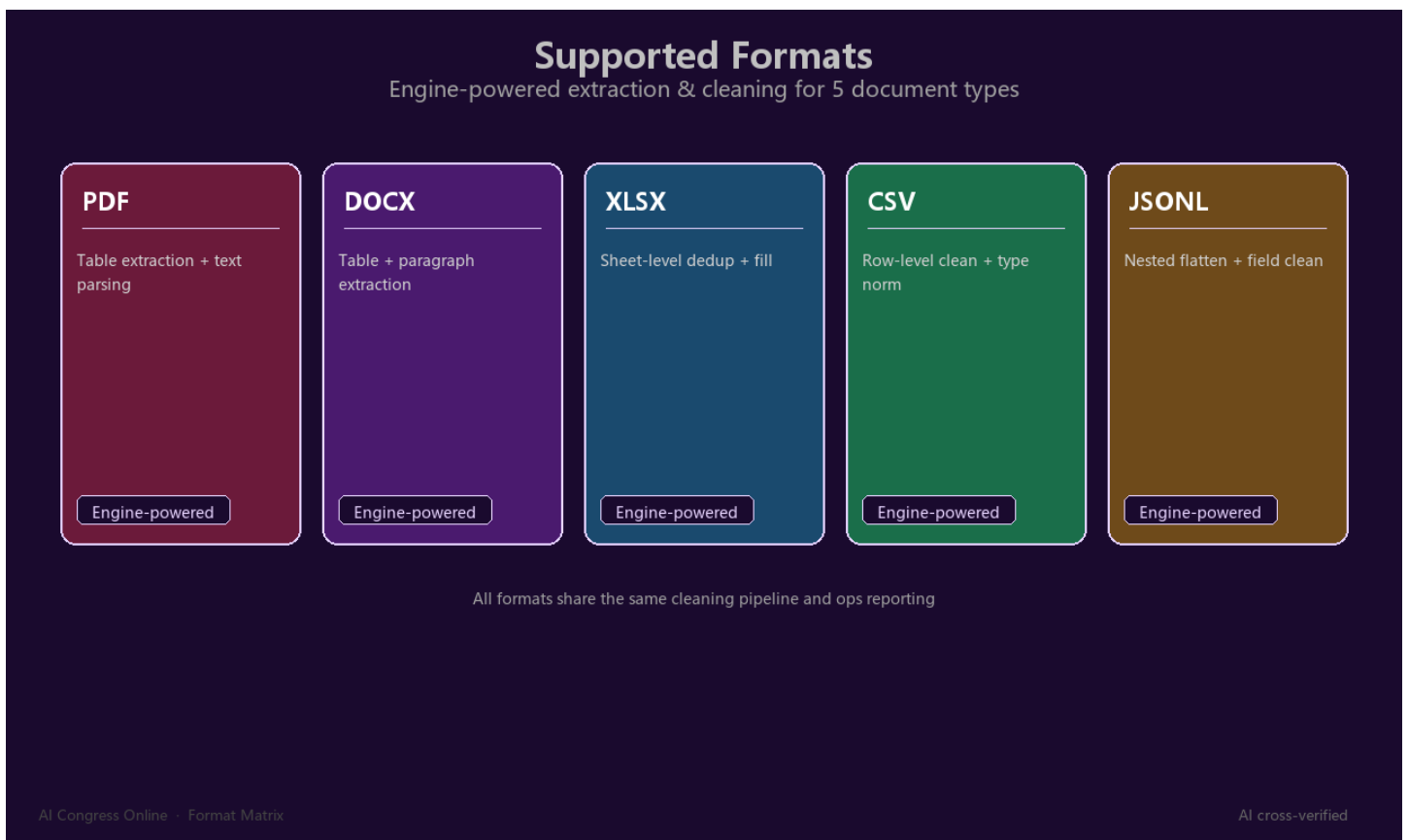
Pipeline, reporting, and format support



Cleaning Report Sample
Detailed ops breakdown for every run

RUN REPORT - 2026-09-18 - job_7f3a9c		
drop_duplicates	-1 row	Removed duplicate customer record by email
strip	2 cells	Trimmed leading/trailing whitespace from name fields
fill_missing	3 cells	Filled empty email and score cells with marker
normalize_type	3 cells	Converted score strings to integers
SUMMARY		
Input: 5 rows Output: 4 rows Ops: 4 Status: CLEAN		
AI cross-verified		

AI Congress Online - Report Engine AI cross-verified



5-Step Pipeline: Ingest -> Parse -> Clean -> AI Verify -> Deliver

Every run produces a detailed ops report (drop_duplicates, strip, fill_missing, normalize_type) with affected row counts. Supports PDF, DOCX, XLSX, CSV, and JSONL formats through the same engine.

Ready to clean your data?

Visit aicongressonline.com or contact on Fiverr: [xinjin376](#)

AI Congress Online - Engine-powered - Cross-verified - Production-ready